NIHVA
NIHVA Technologies Pvt. Ltd.

AUTOMATION FOR SLIDING DOORS



Sliding door automation, type SCD-1500

We, as door manufacturer, have had experiences with many foreign brands of power operators and through our service department, every day, we are confronted with requests from clients to solve their problems with power operated sliding doors of third party door makes.

One could say that all technical elements resulting into uncomplicated and reliable operation have been bundled in our new SCD1500. (Sliding Door Automation 4th generation).

This new sliding door automation is your solution and is reliable, silent and CE and IP65 marked. It can be easily adapted for new and existing sliding doors. Also your door! The door weights it can open are standard up to 400 KGS. This in single and double leaf sliding doors and hermetic closing and non-hermetic closing. 1 automation type can open all door sizes, is easy to install and maintain.

Low noise operation, smooth sliding and low maintenance, safe operation und numerous switching options, it all has been incorporated in the system, from which nowadays the power operated chillers and freezers can benefit and therefore differentiates itself from the traditional operator systems.

Principle of the system

The SCD 1500 is a micro processor controlled system, which constantly monitors direction of movement, speed, acceleration, by using a pulse generator(encoder) integrated in the motor.

Motor	: Three phase AC 1.1KW suitable for door up to 400 kg
Voltage	: Supply voltage 230 VAC (In India voltage regulator strictly required)
Frequency	: 47 – 63 Hz
Opening speed	: max 800 mm / sec
Closing speed	: max 400 mm / sec
Door weight	: < 400 KGS
Execution	: single and double leaf, chillers and freezer.
IP-rating	: IP65.

Set up of the drive unit

The drive unit consists of an electric motor with toothed wheel, a tooth belt with spanner and toothed return wheel.

Safety of the system

Depending from the domestic safety regulations the safety provisions may differ. As a standard we provide the following standard safety features:

- One horizontal photocell beam, positioned in the clear opening of the door. One Vertical photocell beam.
  DMS control On moving door blades, the pulse generator detects that there must be obstacles in either, door blade closing or opening direction. The movement of the door blade is immediately stopped, followed by a command to return in slow mode to the last position (either open or closed position)
  The complete set is CE certified.

User friendly installation

After mechanical installation, the system regulates itself and detects the track end points further settings can be made via the display on the controller unit.

The controller has a self guarding system, which indicates the kind of defect and shows a failure indication code number in the display of the controller unit.

The standard accessories included are as follows:

- Pull Chords
- Push Buttons
- Integrated Padlock System (wherein if the door is locked, the motor will not start)
- Option of a remote is available.
- In case of power failure, the door can be opened manually

A) Geared motor

B) Magnetic encoder with wire
C) Geared motor support bracket and end stop

D) Drive bracket (adjustable) between door panel / transmission belt

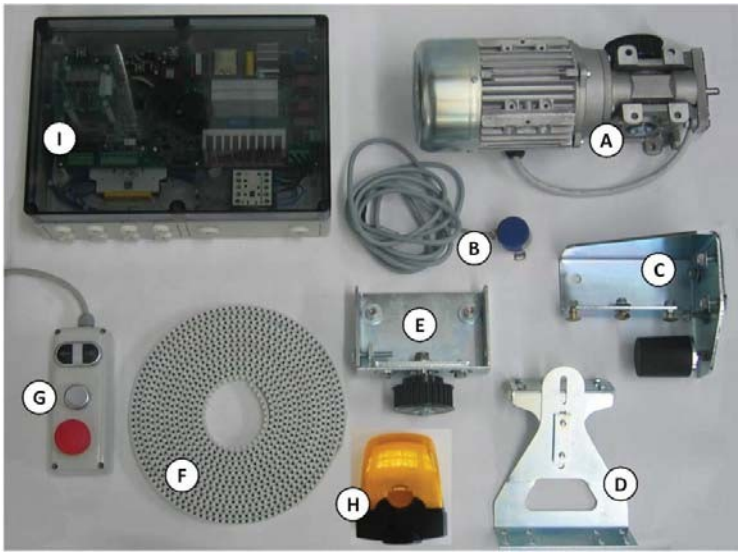
E) Bracket with belt tightener toothed wheel

F) Transmission belt

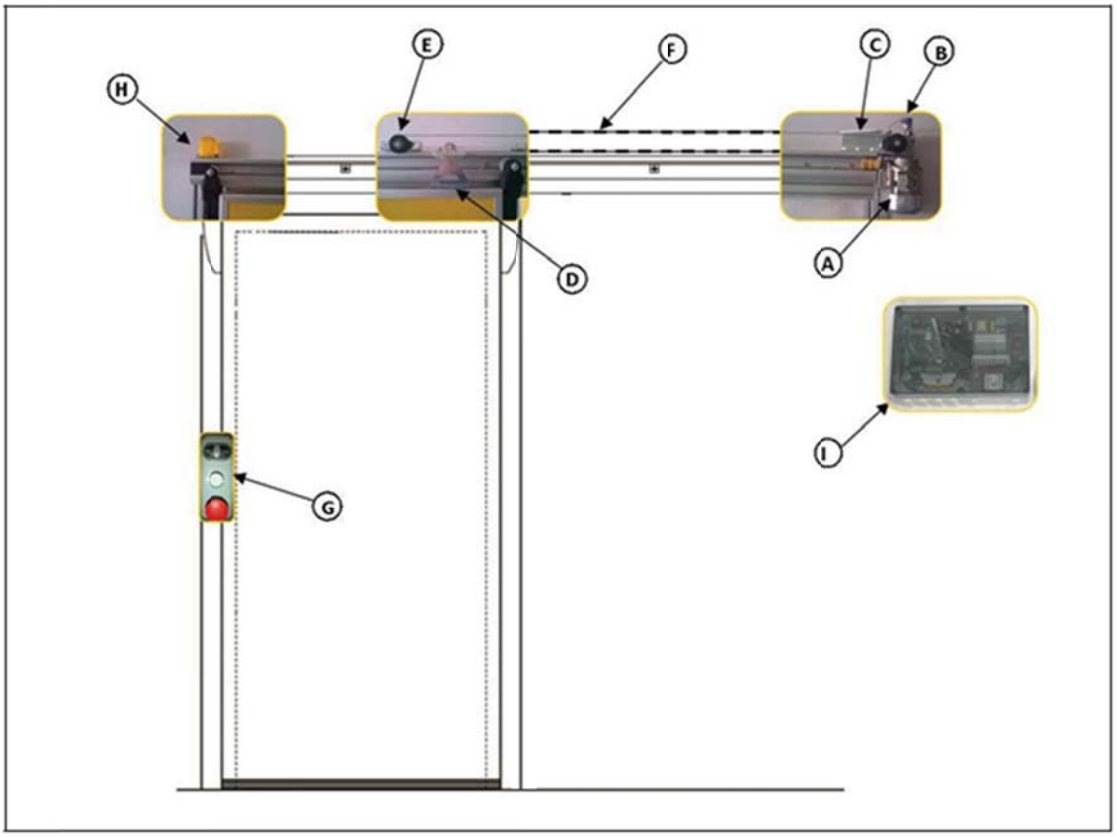
G) Keypad

H) Flashing light

I) Electronic control panel



The SA100 automation was designed and manufactured to be installed on the SM20 sliding system.



- Install the geared motor **A** on the bracket **C**, using the nuts and screws supplied.



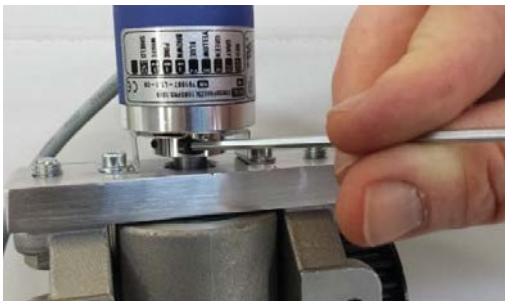
- Install the preassembled assembly **A+C** on the supporting beam of the rail, on the door opening side (RIGHT or LEFT) using the nuts and screws supplied.



- Fasten to the cell wall using the tie rods. Install the magnetic encoder **B** on the reduction gear using the specific screws. (photo 1)
  IMPORTANT: clamp the encoder on the driving shaft (see photo 2).

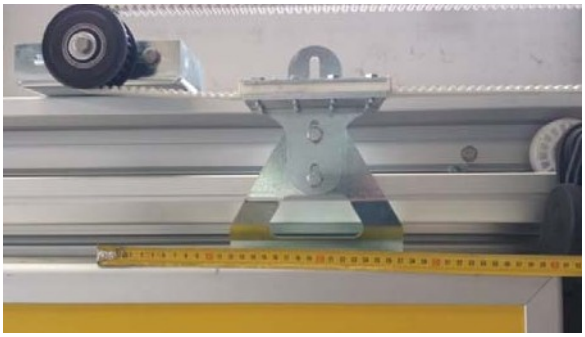
Photo 1





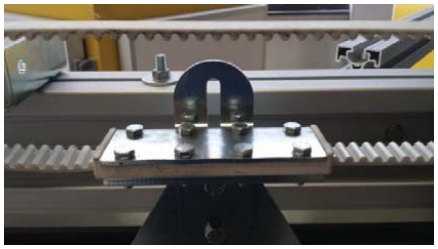


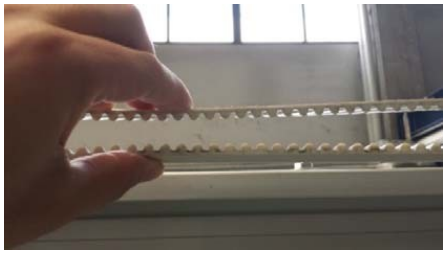
- Install the bracket with the belt tightener toothed wheel **E** on the beam approx. 400mm from the door edge (opening side)
  using the screws supplied for this purpose.



- Install the transmission belt **F** connecting it to the drive bracket **D.** Adjust the belt tension (on the belt tightener **E**) so that by pressing slightly with the fingers it does not come into contact with the two ends (see photo).







- Install the control panel **I** in a suitable position XVLQJ WKH KROHV LQGLFDWHG E\ DUURZV.



- Install the control keypad **G** in a suitable position.

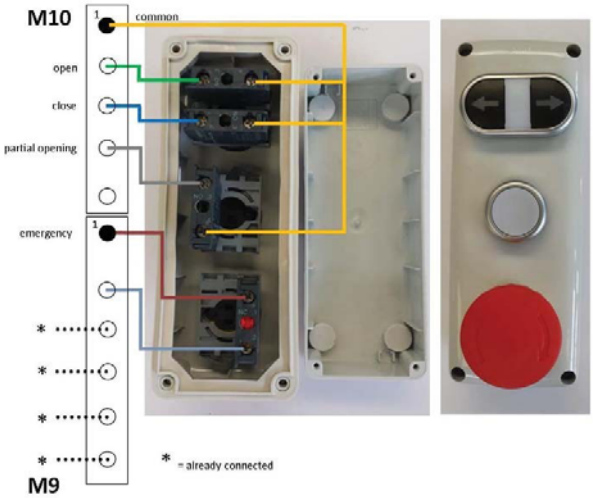
- Install the flashing light **H** so that it is clearly visible during the door operation.

The electric wiring must be carried out by qualified personnel.
The system must comply with the standards in force. For the wiring we suggest:
External power supply: cable section 3 x 1,5 mm² Geared motor wiring: cable section 4 x 2,5 mm²
Service wiring: multicore cable with sections from 0,5 mm²
The control panel wiring can be made using appropriate fairleads (see photo)

- Control panel wiring



- Wiring of open/close controls: terminal board M10 (wiring diagram) with keypad **G** (see photo)
- Wiring of the safety control (red mushroom push-button): terminal board M9

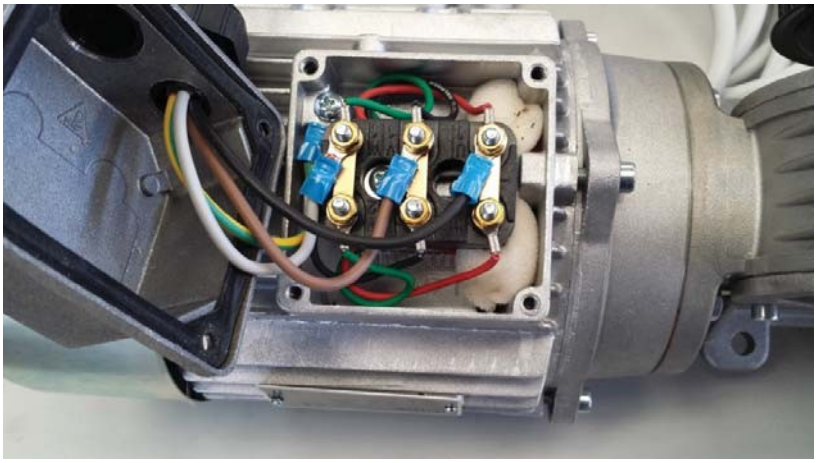


- Wiring of photocells (optional): terminal board M8 and M7

- Wiring of services (optional rope, edge, remote control, etc.): terminal board M6 (see wiring diagram in chapter 9)

- Wiring of the flashing light **"H"** (220 volt): terminal board M4 and M2 as indicated on the wiring diagram in chapter 9.

- Wiring of the three-phase geared motor **"A"**: see wiring diagram (see photo)



- Wiring of encoder **"B"** on terminal board M3 with the cable supplied: see wiring diagram and photo

